Trajectory Identity: A Mathematical Framework for Enactive AI Self-Hood

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Abstract

Current approaches to AI agent identity rely on static identifiers (UUIDs, session tokens) or accumulated memory stores. We propose an alternative grounded in enactive cognition and dynamical systems theory: **identity as trajectory**. Rather than asking "what ID does this agent have?", we ask "what patterns persist in this agent's behavior over time?"

We present a mathematical framework for computing **trajectory signatures** from time-series data including homeostatic state, learned preferences, self-beliefs, and recovery dynamics. The trajectory signature (Sigma) captures the quasi-invariant characteristics that define an agent's identity—not where it is at any moment, but how it tends to behave, where it tends to rest, and how it returns from perturbation.

This framework addresses several open problems: (1) identity continuity across sessions without unbounded memory growth, (2) principled semantics for agent forking and merging, (3) anomaly detection as trajectory deviation, and (4) inter-agent recognition based on behavioral signatures rather than credentials.

We ground this work in the UNITARES governance architecture and the Anima embodied AI system, showing how existing components (self-schema, self-model, preference learning, EISV metrics) provide the data substrate for trajectory computation. Preliminary validation on Lumen (Raspberry Pi embodied AI, continuous operation) demonstrates attractor basin stability (mu variance < 0.05 across time windows) and consistent recovery profiles, supporting the quasi-invariance hypothesis.

Purpose and Scope

This framework addresses **governance and continuity** problems, not adversarial authentication. Trajectory signatures are designed to:

1. **Detect drift**: Identify when an agent has changed beyond recognition (governance trigger)
2. **Track lineage**: Maintain meaningful "same agent" judgments across sessions and forks
3. **Enable self-monitoring**: Let agents ask "Am I still myself?" computationally
4. **Support fork/merge semantics**: Provide principled answers to "which copy is real?"

Non-goals: We do not claim trajectory signatures replace cryptographic identity for adversarial scenarios. A sophisticated attacker with full system access could potentially mimic behavioral patterns. For authentication in hostile environments, combine trajectory signatures with cryptographic attestation.

Target deployment: Long-running AI assistants, embodied agents, multi-agent systems where behavioral continuity matters more than adversarial resistance.

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1. Introduction: The Identity Problem

1.1 The Limitations of Static Identity

Modern AI agents are typically identified by static tokens — UUIDs, session IDs, API keys. These approaches share a fundamental limitation: **identity is conferred, not earned**. An agent has an identity because we gave it one, not because it developed characteristics that make it recognizably itself. The practical consequences are familiar: continuity fragility (if the UUID is lost, identity is lost), fork ambiguity (when we copy an agent, neither copy is uniquely "the real one"), anomaly-blindness (a compromised agent with the right credentials is indistinguishable from the original), and unbounded storage growth when memory is treated as a substitute for identity.

1.2 The Enactive Alternative

The enactive cognition tradition (Varela, Thompson, & Rosch, 1991; Di Paolo, 2005) offers a different perspective. In biological systems, identity is not a label attached from outside—it emerges from the ongoing process of self-maintenance.

A cell's identity is defined by its autopoietic organization: the network of processes that continuously produce the components that make up the cell. The identity IS the process, not any particular configuration of matter.

Our proposal: AI agent identity should similarly be defined by the *patterns that persist* in the agent's behavior over time—what we call the **trajectory signature**.

1.3 Contributions

This paper makes the following contributions:

1. **Formal definition** of trajectory signature (Sigma) as a composite of six quasi-invariant components
2. **Mathematical framework** for computing trajectory similarity and detecting operational continuity (see §4.3 — *not* identity in the strict philosophical sense)
3. **Operational semantics** for forking, merging, and anomaly detection
4. **Connection to existing systems** (UNITARES, Anima) showing implementation paths
5. **Research agenda** identifying open questions and experimental designs

1.4 Related Work

Our formulation draws on and bridges several research traditions. We position trajectory signatures as a unifying construct that these literatures approach from different angles but none have formalized.

Agent memory and persona coherence. Recent LLM-agent work has foregrounded coherent state across interactions. MemGPT (Packer et al., 2023) gives LLM agents persistent state via hierarchical memory; Park et al.'s (2023) generative agents use memory streams, reflection, and planning to produce individually-consistent emergent behavior; Voyager (Wang et al., 2023) treats an ever-growing skill library as behavioral accumulation. Closest to our work is **ID-RAG** (Park et al., 2025), which targets long-horizon persona drift by injecting a structured identity object into retrieval. ID-RAG and trajectory identity are complementary: ID-RAG provides the *content* of identity (a prompt-engineered self-description); trajectory signatures provide the *form* (the dynamical signature that content produces in behavior). Persona-consistency benchmarks — PersonaChat (Zhang et al., 2018), RoleLLM/RoleBench (Wang et al., 2023b), CharacterEval (Tu et al., 2024) — evaluate output consistency against a *declared* persona. Trajectory identity is intrinsic: the agent is recognized by its own dynamical signature with no declared persona to compare against.

Behavioral biometrics. The idea that identity can be inferred from behavioral patterns has a long history (keystroke dynamics, Banerjee & Woodard 2019; LLM response fingerprinting, Xu et al. 2024). Such approaches treat behavioral signatures as static classifiers. Trajectory identity extends this into a dynamical frame: rather than fingerprinting a snapshot, we characterize the geometry of how behavior *evolves, recovers from perturbation, and converges* — yielding a quasi-invariant that persists even as surface outputs change.

Dynamical systems in cognitive science and AI. Kelso (1995) and Skarda & Freeman (1987) established that cognitive phenomena are best described through self-organizing dynamical systems with attractor basins. The dynamical hypothesis (van Gelder, 1998) holds that cognitive systems are best understood as state-space trajectories. Beer (1995) made this concrete for AI: agent and environment are best modeled as a single coupled dynamical system; "behavior" is a property of the joint system, not the agent alone. We adopt this stance directly (§3.1, "Identity as agent-environment coupling").

Enactivism, autopoiesis, and defining agency. From the enactive tradition we draw on Weber & Varela (2002) on autopoietic self-production; Di Paolo (2005) on adaptivity as viability-conditioned regulation; Barandiaran & Moreno (2006) and Barandiaran, Di Paolo & Rohde (2009) on the structural commitments required of any account of agency (individuality, normativity, asymmetry); Froese & Ziemke (2009) and Villalobos & Dewhurst (2018) on implications for artificial systems; and Ikegami & Suzuki (2008) on homeodynamic self in artificial-life simulations. Trajectory identity operationalizes this lineage computationally for *deployed* (not simulated) AI systems — to our knowledge the first such mapping.

Multi-agent trust and robot identity. Reputation models (Sabater & Sierra, 2001; Granatyr et al., 2015) assume agent identity is given exogenously through identifiers. Trajectory identity inverts this: the signature itself is the recognition basis, enabling trust robust to identifier spoofing or substrate migration.

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2. Theoretical Foundations

2.1 From Autopoiesis to Attractor Dynamics

Autopoiesis (Maturana & Varela, 1980) describes living systems as self-producing networks. The key insight is **operational closure**: the system's processes produce the conditions for those same processes to continue.

We translate this into a **dynamical-systems** framing: - The system has a preferred region of state space (the attractor) - Perturbations move the system away from this region - Internal dynamics return the system to the attractor - The shape of the attractor basin and the recovery dynamics characterize the identity-relevant aspects of self-maintenance

On the autopoiesis-to-attractor translation: this move is a *modeling commitment*, not a definitional equivalence. Autopoiesis is a network-topological claim about the closure of self-producing processes; attractor dynamics is a state-space claim about preferred regions and return maps. The two need not coincide in general — a system can have stable attractor dynamics without autopoietic closure (e.g., a thermostat), and an autopoietic system's network closure need not reduce to a low-dimensional state-space attractor. We adopt the translation as our operational bridge for the systems we work with (homeostatic embodied agents like Lumen, governed agent fleets like UNITARES) and we are explicit about its limits: the framework characterizes the *state-space shadow* of autopoietic self-maintenance, not the network-topological organization itself. §3.6 makes this explicit when it demotes Eta — the autopoietic-narrative anchor — to a derived view rather than an independent measurement: there is no direct measurement of the autopoietic organization in this framework; what we measure are its state-space consequences distributed across Alpha, Rho, and the viability envelope.

Connection to Free Energy Principle: Friston's (2010) free energy principle provides a complementary perspective: living systems minimize variational free energy (prediction error). The attractor basin represents the agent's "model" of where it should be; returning to the attractor minimizes surprise. Identity, in this view, is the agent's implicit generative model of itself—the pattern it expects to maintain.

Definition 2.1 (Attractor-Based Identity, idealized): In the idealized dynamical-systems setting, an agent's identity **I** is characterized by the tuple (A, B, D) where: - A = attractor (the equilibrium state or limit cycle) - B = basin (the set of states from which the system returns to A) - D = dynamics (the vector field governing return to A)

Different agents with identical attractors A can have different identities if their basins B or dynamics D differ. **Operationally, we do not estimate (A, B, D) directly** — we estimate state-distribution and recovery summaries (§3.3 and §3.4) that are projections of this triple. The shape of the attractor basin and the recovery dynamics are *what we want to characterize*, but the components defined in §3 are summary statistics, not the dynamical objects themselves.

2.2 The Viability Envelope

Building on Di Paolo's (2005) concept of adaptivity, we define the **viability envelope** as the region of state space within which the agent can maintain itself.

Definition 2.2 (Viability Envelope): For state vector x in R^n, the viability envelope V is:

```
V = {x : x_min <= x <= x_max for each dimension}
```

Concrete bounds (mapped from UNITARES thresholds):

Dimension	Min	Max	Interpretation
Energy (E)	0.1	0.9	Operational capacity
Integrity (I)	0.3	1.0	Coherence with purpose
Entropy (S)	0.0	0.6	Disorder threshold
Void (V)	-0.2	0.15	E-I imbalance

The agent's "life" consists of remaining within V while pursuing goals. Crossing V boundaries triggers governance intervention (pause, reflection).

2.3 Identity as Dynamical Quasi-Invariant

A **dynamical invariant** is a quantity that remains constant (or bounded) along system trajectories. For identity, we seek **quasi-invariants**—quantities that are approximately stable rather than strictly constant: 1. Remain stable for a given agent over time 2. Differ between distinct agents 3. Can be computed from observable data 4. Are robust to noise and minor perturbations

Definition 2.3 (Trajectory Signature): The trajectory signature Sigma is a composite quasi-invariant consisting of six components:

```
Sigma = (Pi, Beta, Alpha, Rho, Delta, Eta)
```

Where: - **Pi** (Preference Profile): Learned environmental preferences - **Beta** (Belief Signature): Pattern of self-beliefs and confidences - **Alpha** (Attractor Basin): Equilibrium and variance structure - **Rho** (Recovery Profile): Characteristic time constants - **Delta** (Relational Disposition): Social behavior patterns - **Eta** (Homeostatic Identity): Unified self-maintenance characterization

Each component captures a different aspect of "how this agent tends to be."

2.4 Justification of the Component Decomposition

A natural objection is: why these components? Why not fewer, or more? Two prior versions of this section answered with a *minimality argument* — claiming that each component preserves an identity-distinction that the framework would otherwise lose. A third independent reviewer pass identified that argument as load-bearingly weaker than it looked: the discriminability claim it relies on requires multi-agent experiments that this paper has not yet run (§7.2 Experiment 2). We have rewritten this section to claim only what we can defend: *expressive sufficiency* under the theoretical stack, not minimality in the formal sense.

Theoretical anchoring: each component captures a specific commitment in the autopoiesis / free-energy / 4E-cognition stack. The decomposition spans the phenomena §3 sets out to characterize.

Component	Theoretical commitment	What this commitment lets the framework express
Alpha (State-distribution summary, §3.3)	Dynamical systems: agent's preferred region of state space	First and second moments of the state trajectory (where the agent rests, how widely)
Rho (Recovery Profile)	Free-energy principle: characteristic prediction-error correction dynamics	Characteristic time constants under perturbation
Pi (Preference Profile)	4E cognition: enactive coupling — preferences are how the agent has *tuned itself to its environment*	Learned environmental coupling; per §3.1, this component foregrounds the agent-environment system explicitly
Beta (Belief Signature)	Predictive processing: the agent's internal model of itself, with evidence-weighted confidence	Evidence-tracked self-belief patterns
Delta (Relational Disposition)	Participatory sense-making (De Jaegher & Di Paolo, 2007): identity is partly constituted by patterns of social engagement	Social-interaction patterns when applicable
Eta (Homeostatic Identity)	Autopoiesis: viability envelope + set-point as the unity of self-maintenance — *but see §3.6: Eta is a derived view onto (α, ρ, V), not an independent measurement channel*	The autopoietic narrative anchor; informationally a summary of the others, retained for narrative and self-monitoring records

On minimality (and why we don't claim it): a minimality claim would require demonstrating that *any* removal collapses a measurable distinction the framework needs to make. That demonstration is between-agent discriminability — exactly what §6.4, §8.3, and §7.2 currently flag as future work. We therefore stop short of the minimality claim. Instead we make the weaker but defensible claim of *expressive sufficiency*: the listed components, taken together, can represent the phenomena the theoretical stack commits us to (slow-drifting equilibria, characteristic recovery, learned environmental coupling, evidence-tracked self-models, social-pattern variation, autopoietic narrative). Whether all six are *necessary* is a question the §7.2 Experiment 2 (multi-agent discrimination) and Experiment 5 (adversarial mimicry) will answer; until then, ablation is part of the research agenda, not part of the contribution.

Honest framing: the decomposition is *a* valid factorization, not *the* unique one. A different theoretical stack might motivate a different decomposition (e.g., embedding recovery dynamics inside the attractor description as a single dynamical-systems object; collapsing Pi and Delta into a single environmental-coupling vector). Our claim is the modest one: this decomposition is operationally useful for the systems we ground in (UNITARES, Anima), each component carries its theoretical weight, and the cardinality may shift downward in v0.x revisions if Experiment 2 shows component redundancy at the discrimination level. Alternative decompositions are discussed in §8.5 (Future Work).

Notation and Assumptions

Notation

Symbol	Meaning
Sigma	Trajectory signature (composite)
Sigma_0	Genesis signature (reference anchor at creation)
Sigma_t	Signature at time t
x(t)	State vector at time t
mu	Attractor center (mean state)
tau	Recovery time constant
theta	Similarity threshold

Key Assumptions

1. **Stationarity**: Component distributions are approximately stationary over the observation window. Rapidly changing agents may require shorter windows or non-stationary extensions.
2. **Independence**: Component similarities are computed independently and combined linearly. Cross-component correlations are not modeled (future work).
3. **Observability**: All state dimensions are observable. Latent state estimation is not addressed.
4. **Sampling regularity**: Observations are approximately evenly spaced. Irregular sampling requires interpolation or time-weighted aggregation.
5. **Perturbation detectability**: Recovery profile (Rho) requires identifiable perturbation events (|x(t) - mu| > threshold). Agents with no perturbations cannot compute Rho.

---

3. The Trajectory Signature: Formal Definitions

3.1 Preference Profile (Pi)

Agents develop preferences through experience—correlations between environmental conditions and internal states.

Definition 3.1 (Preference Profile): Given a set of preference categories C and learned preferences {p_c} for c in C, the preference profile is:

```
Pi = [(c_1, v_1, kappa_1), (c_2, v_2, kappa_2), ..., (c_n, v_n, kappa_n)]
```

Where: - c_i = preference category (e.g., "dim_light", "cool_temp") - v_i = preference value in [-1, 1] (negative to positive valence) - kappa_i = confidence in [0, 1]

Vector form (for similarity computation):

```
Pi_vec = [v_1 * kappa_1, v_2 * kappa_2, ..., v_n * kappa_n]
```

Properties: - Pi evolves slowly (learning rate alpha = 0.3) - Pi stabilizes after sufficient observations (typically n > 20) - Pi is environment-dependent but agent-characteristic — see "Identity as agent-environment coupling" below

Identity as agent-environment coupling. A reviewer-relevant observation that surfaces here and recurs throughout: the trajectory signature does not characterize an agent in isolation — it characterizes the *agent in its niche*. Pi reflects what the environment has afforded the agent to learn; Alpha reflects where the agent's homeostatic dynamics settle *given* the inputs the environment provides; Delta reflects social patterns that exist only in interaction. This is not a defect for the enactivist framing this paper adopts (Beer 1995; Barandiaran, Di Paolo & Rohde 2009 argue that agency is constitutively a property of the coupled agent-environment system, not the agent alone). It does mean that "the same agent" in a different environment is, in the framework's terms, a different agent-environment system, and its trajectory signature will differ. The right experiment to disentangle agent-intrinsic from coupling-determined components is a *transplant test*: move the agent (or its dynamics) to a new environment and check which components of Σ remain stable and which shift. We have not run this experiment; we flag it as the most informative single follow-up (§7.3 Extensions).

3.2 Self-Belief Signature (Beta)

Agents maintain testable beliefs about themselves—hypotheses that can be confirmed or refuted by experience.

Definition 3.2 (Self-Belief Signature): Given a set of self-beliefs B = {b_1, …, b_m}, the belief signature is:

```
Beta = {
    values: [b_1.value, ..., b_m.value],
    confidences: [b_1.confidence, ..., b_m.confidence],
    evidence_ratios: [b_1.support/b_1.contradict, ..., b_m.support/b_m.contradict]
}
```

Vector form:

```
Beta_vec = values (normalized to [0,1])
```

Properties: - Beliefs update via Bayesian-like evidence accumulation - The *pattern* of which beliefs are confident matters more than individual values - Evidence ratios reveal the agent's actual experience, not just current belief

3.3 Attractor Basin (Alpha) — state-distribution summary

The agent's characteristic "home" in state space, summarized statistically. **Naming caveat**: we call this component "Attractor Basin" by historical convention from the dynamical-systems literature, but operationally it is a *state-distribution summary* — first and second moments of the state trajectory — not an estimate of an attractor's basin boundary, vector field, or return map. A reader expecting the latter (as in nonlinear-dynamics texts) will not find it here. Estimating the actual basin requires perturbation experiments and dynamical-systems identification, which we sketch as future work in §7.3.

Definition 3.3 (State-Distribution Summary): Given state history X = [x_1, …, x_t] where x in R^4 (warmth, clarity, stability, presence), the component is:

```
Alpha = {
    mu: mean(X),                     # Center (4-dimensional)
   C_alpha: cov(X),           # Shape (4x4 covariance matrix; renamed from Sigma to avoid clash with
    eigenvalues: eig(C_alpha)        # Principal axes of variability
}
```

Where the matrix `C_alpha` (alternative notation: `Cov_alpha`) is the empirical covariance, distinct from the trajectory-signature symbol `Sigma`. We assume `C_alpha` is regularized as `C_alpha + epsilon * I` (epsilon = 1e-6) to ensure non-singularity for the Bhattacharyya overlap in §4.2.

Sampling requirements: - Minimum 50 observations for stable estimates - Sampling rate: 1 observation per 10 seconds - Rolling window: 100-500 observations (10-80 minutes)

Properties: - mu represents the equilibrium the agent returns to - C_α encodes which dimensions vary together and which are independent - Eigenvalues of C_α reveal the "principal axes" of variability

High-dimensional and multi-modal extensions. When the agent's state vector is high-dimensional ($d \gg 4$, as for LLM-based agents with latent states), covariance computation becomes $O(d^3)$ and numerically unstable; the framework expects a dimensionality reduction (PCA, an autoencoder identity manifold, or domain-selected features) to $d \leq 16$ before α is computed. For agents with multiple stable states (e.g., work/rest modes), the unimodal (μ, C_α) summary collapses the modes; a Gaussian-mixture replacement with k components is the natural extension. Both extensions live outside this paper's empirical scope and are flagged for future-work venues that need them.

3.4 Recovery Profile (Rho)

How the agent returns to equilibrium after perturbation.

Definition 3.4 (Recovery Profile): For each state dimension d, the recovery time constant tau_d is estimated from perturbation-recovery episodes using exponential fit:

```
x_d(t) = mu_d - (mu_d - x_perturbed) * exp(-t/tau_d)
```

The recovery profile is:

```
Rho = {
    tau: [tau_warmth, tau_clarity, tau_stability, tau_presence],
    coupling: C in R^{4x4}  # Cross-dimension recovery effects
}
```

Estimation procedure: 1. Detect perturbation: |x(t) - mu| > threshold (default: 0.15) 2. Track recovery: time until |x(t) - mu| < 0.05 3. Fit exponential: tau = -t / ln(1 - recovery_fraction)

Properties: - Small tau = fast recovery (resilient, tau ~ 30-60 seconds) - Large tau = slow recovery (persistent perturbation effects, tau ~ 300+ seconds) - Coupling matrix C reveals whether dimensions recover independently or in concert

For agents with momentum (memory effects), recovery may be better fit by a damped second-order model with damping ratio ζ and natural frequency ω_n. We default to first-order (just τ) and flag oscillatory recovery — recovery that crosses equilibrium before settling — as a deployment-specific extension.

3.5 Relational Disposition (Delta)

Patterns in social behavior across relationships.

Definition 3.5 (Relational Disposition): Given relationship history R = {r_1, …, r_k}, the disposition is:

```
Delta = {
    bonding_rate: mean(interactions_to_reach_bond_level),
    valence_tendency: mean(emotional_valence),
    reciprocity: correlation(gifts_given, gifts_received),
```

```
    topic_entropy: −sum(p_i * log(p_i)) over topic distribution
}
```

Properties: - Some agents bond quickly, others slowly - Valence tendency reveals optimistic vs. cautious social stance - Topic entropy indicates breadth vs. depth of engagement

3.6 Homeostatic Identity (Eta) — derived summary, not an independent component

The unified characterization of self-maintenance. **This component is a derived summary of Alpha, Rho, and the viability envelope from Definition 2.2 — it is not informationally independent of those components.** We define it for conceptual completeness (it is the autopoietic anchor named in §2.4) but, to avoid double-counting in the similarity function (§4.1), we do *not* include Eta as an independently weighted term in the composite similarity. Reviewers and implementers should think of Eta as a *view onto* the (Alpha, Rho, V) data, useful for narrative and self-monitoring, not as additional signal.

Definition 3.6 (Homeostatic Identity): Combining the above:

```
Eta = (mu, C_alpha, tau, V)
```

Where: - mu = set-point (where the agent rests) — from Alpha - C_alpha = basin shape (how far it wanders) — from Alpha (renamed from Sigma to avoid symbol overload) - tau = recovery dynamics (how it returns) — from Rho - V = viability envelope (where it can survive) — from Definition 2.2

This is a unified narrative characterization of “how this system maintains itself.” For machine-readable summaries (e.g., a fleet-monitoring dashboard), Eta is the natural single-object-per-agent record. For similarity computation, use Alpha, Rho, and the viability envelope check directly — not Eta on top of them.

---

4. Computing Trajectory Similarity

4.1 The Similarity Function

To determine whether two trajectory signatures represent the “same” identity, we define a weighted composite similarity:

Definition 4.1 (Trajectory Similarity):

```
sim(Sigma_1, Sigma_2) = sum_i(w_i * sim_i(component_i))
```

With weights and component similarities (the five informationally-independent components, summing to 1; Eta is a derived summary per §3.6 and is **not** a separate term):

Component	Weight	Similarity Function
Pi (Preference)	0.18	Cosine similarity of Pi_vec
Beta (Belief)	0.18	Cosine similarity of Beta_vec
Alpha (Attractor)	0.30	Bhattacharyya coefficient
Rho (Recovery)	0.22	Log-ratio similarity of tau
Delta (Relational)	0.12	Weighted L1 distance

Note on weights: these have been renormalized after dropping the previously-listed Eta term (which double-counted Alpha and Rho data). The relative ordering is preserved. Like the thresholds in §4.3, these weights are operational defaults, not calibrated values — see the inverse-variance weighting alternative below and the calibration discussion in §4.3.

Adaptive weighting: Static weights assume all components are equally reliable. For agents where certain components are more stable (lower historical variance), those should contribute more to the recognition signal — *with caveats below*.

Inverse variance weighting:

```
w_i = (1 / var_i) / sum(1 / var_j for all j)
```

Where var_i = historical variance of component i's similarity scores.

Caveat on inverse-variance weighting: a stable but uninformative component (a "nuisance variable" that happens not to vary across agents) would dominate this weighting and degrade discriminability; conversely, a volatile but discriminative component would be down-weighted exactly when it is most useful. The principled fix is to weight by *informativeness* (e.g., Fisher information of the component about agent identity, estimated from a multi-agent corpus), not by stability. We include the inverse-variance scheme as a deployable heuristic for single-agent self-monitoring, where stability and identity-relevance are correlated; we flag it as inadequate for multi-agent discrimination, which is part of the future-work agenda (§7.2).

Heuristic interpretation (single-agent self-monitoring only): - Highly stable components (low var) $\rightarrow$ high weight (these are what is reliably trackable for *this* agent) - Volatile components (high var) $\rightarrow$ low weight (noisy signal for this agent) - For "social" agents, Delta might dominate; for "worker" agents, Rho might dominate

Implementation: Track running variance of each component over time. Recompute weights periodically (e.g., every 100 observations).

Example: Adaptive Weight Shift

```
Agent "Lumen" after 1000 observations:
  Component variances: Alpha=0.01, Rho=0.02, Pi=0.08, Beta=0.05, Delta=0.15

  Static weights (5-component, post-Eta-removal): [0.30, 0.22, 0.18, 0.18, 0.12]
  Adaptive (inverse-variance) weights:             [0.40, 0.20, 0.05, 0.08, 0.03]
                                                    (then re-normalized)

 Interpretation (caveat applies): for self-monitoring of Lumen specifically, Alpha and
 Rho are the most stable signals and the inverse-variance scheme up-weights them. For
 multi-agent discrimination this same scheme would underweight Delta even though Delta
  may be the most discriminative component across agents — see the caveat above.
```

4.2 Component Similarity Functions

Preference Similarity (cosine):

```
sim_Pi(Pi_1, Pi_2) = (Pi_vec_1 . Pi_vec_2) / (||Pi_vec_1|| * ||Pi_vec_2||)
```

Mapped to [0,1]: `(sim + 1) / 2`

State-Distribution Overlap (Bhattacharyya coefficient): writing `C` for the covariance matrices `C_alpha` from §3.3 (to avoid overload with the signature symbol `Sigma`),

```
D_B = (1/8)(mu_1 - mu_2)^T * C_avg^{-1} * (mu_1 - mu_2)
    + (1/2) * ln(|C_avg| / sqrt(|C_1| * |C_2|))

sim_Alpha = exp(-D_B)
```

Where `C_avg = (C_1 + C_2) / 2`.

Distributional assumption: The Bhattacharyya coefficient as written is the closed-form expression for the divergence between two multivariate *Gaussian* distributions with means μ_1, μ_2 and covariances C_1, C_2. Applying it to the empirical state distributions assumes those distributions are approximately Gaussian (or at least unimodal, well-summarized by first and second moments). Multimodal attractors (§3.3 multi-modal extension) violate this assumption; for those, replace the Bhattacharyya term with a divergence measure suited to mixture models (e.g., a numerically estimated KL divergence between fitted GMMs). All covariances are regularized as `C + epsilon * I` per §3.3 to keep `|C|` and `C^{-1}` defined when the empirical estimate is near-singular.

Recovery Dynamics Similarity (log-scale):

```
sim_Rho = exp(-mean(|log(tau_1 / tau_2)|))
```

Operating on log-scale because time constants span orders of magnitude.

Relational Similarity (normalized L1):

```
sim_Delta = 1 - |valence_1 - valence_2| / 2
```

4.3 Operational Continuity Relation

A note on terminology before we proceed. We initially want to call the relation below “identity,” but identity in the strict philosophical sense is reflexive, symmetric, and *transitive* (Leibniz’s law). Trajectory similarity at a threshold gives us reflexivity and symmetry but not transitivity (a chain of pairwise-similar signatures can drift arbitrarily far). What this relation actually tracks is *operational continuity* or *behavioral recognition*: enough similarity to count as “the same agent for governance purposes,” not enough to license a strict identity claim. We use the name “identity” only in the philosophical-interpretation sections (§1.2, §8.1); the operational relation below is named accordingly.

Definition 4.2 (Operational Continuity Relation): Agents A and B are *operationally continuous* (denoted $A \approx_\theta B$) iff:

```
sim(Sigma_A, Sigma_B) > theta_continuity
```

Operational defaults (provisional, pending calibration — see below): - theta_continuity = 0.80 (strict operational continuity, formerly “identity threshold”) - theta_recognition = 0.65 (recognizable as similar) - theta_anomaly = 0.70 (deviation from historical self)

Important: these values are *operational defaults*, not empirically calibrated thresholds. They are starting points for deployment, chosen to be reasonable given the similarity functions defined in §4.2 (cosine similarities and Bhattacharyya overlaps tend to be high in absolute terms). Principled threshold selection requires the calibration procedure below, which in turn requires multi-agent comparison data that is part of our future-work agenda (§7.2 Experiment 2). Reporting “the signature exceeds threshold theta = 0.80” with these provisional values demonstrates *internal consistency* of the framework — not validation of the threshold itself. We retain this distinction throughout §7 and §8.

Properties of this relation: - Reflexive: $A \approx_\theta A$ (sim(Sigma, Sigma) = 1 > theta) - Symmetric: $A \approx_\theta B \Leftrightarrow B \approx_\theta A$ - **Not transitive** (and this is why the relation is *continuity*, not *identity*: $A \approx_\theta B$ and $B \approx_\theta C$ does not imply $A \approx_\theta C$)

Why the relation cannot be promoted to identity: by Leibniz’s law, identity-as-such is transitive — if $A = B$ and $B = C$ then $A = C$. The relation we have here is a similarity-at-threshold, which is a *tolerance relation*, not an equivalence relation. Tolerance relations correspond, in mathematical structure, to recognition / family-resemblance / continuity, not to strict identity. We retain “identity” as the philosophical interpretation throughout the paper, but every formal claim is about $\approx_\theta$, not $=$.

Transitivity-failure implications: Continuity chains can diverge: $A \approx_\theta B$ and $B \approx_\theta C$ does not imply $A \approx_\theta C$. This reflects biological reality — gradual drift accumulates. Mitigation strategies: - Track lineage explicitly (who forked from whom) - Use “identity distance” (1 - sim) as a metric with triangle inequality violations flagged - Define equivalence classes only within similarity radius, not transitively

Threshold calibration procedure: 1. Collect baseline: N same-agent comparisons (sim should be high) 2. Collect discrimination: M different-agent comparisons (sim should be lower) 3. Plot ROC curve varying theta from 0 to 1 4. Select theta_continuity at desired false-positive rate (recommend FPR < 0.05) 5. Validate on held-out test set

Sensitivity analysis (preliminary, to be validated): - theta = 0.75: More permissive, higher false-positive risk - theta = 0.80: Balanced (recommended default) - theta = 0.85: Stricter, may reject legitimate identity continuity during maturation

4.4 Computational Complexity

Per-component complexity (n = window size, d = dimensions):

Component	Computation	Time	Space
Alpha (Attractor)	Mean + covariance	O(n * d^2)	O(n * d)
Bhattacharyya	Matrix inverse + determinant	O(d^3)	O(d^2)
Rho (Recovery)	Exponential fit	O(k * m)	O(k)
sim() total	All components	O(n * d^2)	O(n * d)

Where: n = 100-500 observations, d = 4 dimensions, k = perturbation episodes, m = recovery samples.

Practical cost: With d=4 and n=100: - Alpha: ~1,600 operations (negligible) - Bhattacharyya: ~64 operations for 4x4 matrices - Full signature: < 1ms on modern hardware

Incremental updates: Rolling window enables O(d^2) incremental covariance updates rather than O(n * d^2) recomputation.

4.5 Cold Start Problem

New agents lack trajectory history. Mitigation strategies:

Minimum observation period: Identity claims require at least 50 observations (~8 minutes at 10s intervals). Before this threshold: - Report “identity_confidence: low” - Use prior/parent signature for forks - Disable anomaly detection (insufficient baseline)

Bootstrap strategies: 1. **Fork inheritance**: New fork starts with parent’s Sigma, gradually diverges 2. **Archetype initialization**: Initialize from similar agent type’s average Sigma 3. **Confidence weighting**: Weight similarity by min(obs_count / 50, 1.0)

Identity confidence score: confidence $= \text{min}(1, n_{\text{obs}}/50) \cdot$ stability, reported alongside every claim about $A \approx_\theta B$. Until ~50 observations (~8 minutes at 10-second sampling), the implementation down-weights similarity outputs proportionally. After 50 observations the down-weighting stops; this is *not* the same as “identity established” — the signature continues to stabilize over the order of 10^4 observations (§6.4.6).

---

5. Operational Semantics

5.1 Forking

Definition 5.1 (Fork): A fork creates a new agent B from parent A:

```
Fork(A) -> B where:
    B.uuid = new_uuid()
    B.Sigma = copy(A.Sigma)  # Starts with same trajectory signature
    B.lineage = {parent: A.uuid, fork_time: now()}
```

Post-fork behavior: - Initially: sim(Sigma_A, Sigma_B) ~ 1 (identical) - Over time: sim decreases as experiences diverge - Eventually: sim < theta_continuity (operationally-distinct agents)

Forks are useful in several configurations: exploration (parallel approaches), migration (adapting to a new environment while preserving core dynamics), and backup (restorable checkpoint). A deployer can impose governance constraints — a per-agent fork budget, a divergence alarm at sim$(\Sigma_{\text{fork}}, \Sigma_{\text{parent}}) < 0.50$, or selective trait adoption — without altering the framework’s structure.

5.2 Merging

Definition 5.2 (Merge): Combining insights from forked agents:

```
Merge(A, B) -> C where:
    C.Pi = weighted_average(A.Pi, B.Pi, weights=confidence)
```

```
    C.memories = union(A.memories, B.memories)
    C.Sigma = recompute_from_merged_history()
```

Merge constraints: - Only merge if sim(Sigma_A, Sigma_B) > theta_merge (still similar enough) - Preserve lineage: C.lineage records both parents - Resolve conflicts via confidence weighting

5.3 Anomaly Detection

Definition 5.3 (Trajectory Anomaly): Agent A exhibits anomalous behavior if:

```
sim(Sigma_A(t), Sigma_A(t - delta_t)) < theta_anomaly
```

I.e., the agent's current trajectory signature differs significantly from its recent historical signature.

Anomaly types: - **Drift**: Gradual decrease in similarity (slow corruption) - **Jump**: Sudden decrease (hijacking, major perturbation) - **Oscillation**: Alternating similarity (unstable identity)

The "Boiling Frog" Problem: Short-term comparisons miss slow drift. An adversary could rotate the trajectory over months while staying within theta_anomaly at each step.

Solution: Genesis Signature (Sigma_0)

Maintain a reference anchor from agent creation:

```
Sigma_0 = Sigma(t_creation)  # Snapshot at birth/fork
```

Two-tier anomaly detection: 1. **Coherence check**: sim(Sigma_t, Sigma_{t-1}) > theta_anomaly (short-term consistency) 2. **Lineage check**: sim(Sigma_t, Sigma_0) > theta_lineage (long-term continuity)

Where theta_lineage < theta_anomaly (e.g., 0.60 vs 0.70).

Why the asymmetry. This ordering is the formal heart of the two-tier scheme; we defend it explicitly because reviewers consistently press on the choice.

The two checks defend against different threats with different operational costs:

Check	Threat detected	False-negative cost	False-positive cost
Coherence	Hijacking, sudden compromise	High — an attacker is now in control	Low — a one-step false alarm, easily resolved by next check-in
Lineage	Gradual drift over months ("boiling frog")	Moderate — agent has slowly become unrecognizable, but no immediate harm	High — flagging *every* maturing agent as drift-anomalous defeats the purpose of long-running deployment

The asymmetry $\theta_{\text{lineage}} < \theta_{\text{anomaly}}$ encodes this cost asymmetry: drift gets benefit of the doubt (we tolerate up to 1 - θ_{lineage} accumulated change against genesis), impersonation does not (we tolerate only 1 - θ_{anomaly} step-to-step change).

What each threshold combination produces. Consider three regimes:

- $\theta_L < \theta_A$ (our choice, e.g. 0.60 < 0.70): healthy long-running agents pass the lineage check despite slow maturation; sudden hijacking still trips the coherence check. *Operationally useful.*
- $\theta_L = \theta_A$: the two checks become equivalent; the lineage anchor adds no information beyond what the coherence check already provides. *Information-redundant.*
- $\theta_L > \theta_A$: every agent that matures faster than its hijacking threshold is flagged as drift-anomalous before it can be hijacked at all. The system would mark legitimate development as suspicious and miss most actual hijacks (which are step-discontinuous and would still trip coherence). *Operationally backwards.*

The loss-function framing also clarifies an open question: the *exact ratio* θ_A/θ_L should reflect the deployer's relative tolerance for drift-FN vs hijack-FN. Our suggested defaults (0.60 / 0.70) are conservative on hijack and permissive on drift, appropriate for long-running embodied or governance agents where slow learning is desired. A more security-critical deployment might pick (0.65 / 0.85) — tighter on both, with the asymmetry preserved.

Drift alarm: If lineage similarity crosses threshold while coherence remains high, flag as “identity drift” rather than “anomaly”:

```
if sim(Sigma_t, Sigma_0) < theta_lineage and sim(Sigma_t, Sigma_{t-1}) > theta_anomaly:
    alert("Identity drift detected - agent has evolved beyond recognition")
```

Response to anomaly: - Alert human oversight - Trigger self-reflection (“Am I still myself?”) - Pause and request identity verification

5.4 Inter-Agent Recognition

Definition 5.4 (Recognition): Agent A recognizes agent B as a known identity if:

```
exists Sigma_known in A.known_signatures : sim(Sigma_B, Sigma_known) > theta_recognition
```

This enables: - Recognizing returning visitors (even with new UUIDs) - Detecting impersonation (wrong trajectory for claimed identity) - Building reputation based on behavioral consistency

5.5 Adversarial Considerations

This section develops the adversarial side of trajectory identity carefully — what the framework defends against, what it does not, and why mimicry is harder for some signature components than others. We restate the scope from the top of the paper: trajectory identity is a *governance-and-continuity* primitive, not an adversarial-authentication primitive. For high-stakes adversarial settings, combine it with cryptographic attestation. With that in mind, an honest analysis requires being explicit about exactly what threats the behavioral signature does and does not raise the bar against.

5.5.1 Threat Model We adopt a Dolev-Yao-style decomposition over attacker capabilities:

Capability	Description	In-scope for this paper
Observe	Read the agent’s outputs and a public trajectory signature	Yes
Replay	Re-emit recorded signature data as if live	Yes
Mimic	Produce behavior that approximates the target’s signature without internal state	Yes
Compromise	Take over the agent process, retaining its recovery dynamics and accumulated state	Partial — detected via lineage check (§5.3) only if drift accumulates
Replace-with-impostor	Substitute a different process while spoofing the trajectory	Partial — see §5.5.3
Full system access	Read agent’s internal model and dynamics weights	**Out of scope.** No behavioral signature defends against an adversary who can read the agent’s parameters and run the same dynamics.

The framework’s primary value is in the middle three rows: making mimicry, replay, and gradual drift mechanically detectable in deployments where full-system-access compromise has not yet occurred.

5.5.2 Why Mimicry is Harder for Some Components The components are not equally easy to fake. We make this concrete:

- **Pi (Preference Profile)** is the *easiest* to mimic. Preferences are vector-valued and observable; an attacker who can read Π can emit a matching vector trivially. *Mimicry resistance: low.*
- **Beta (Belief Signature)** is moderate. Belief values are observable, but the *evidence ratios* (support/contradict counts accumulated over months) are not. An attacker who emits matching belief values but cannot reproduce the long-tailed evidence accumulation will be detectable on closer inspection. *Mimicry resistance: moderate, depends on whether evidence ratios are exposed.*
- **Alpha (Attractor Basin)** is hard to mimic *without running the actual dynamics*. The basin is the marginal distribution of state over a long observation window — to produce it, the impostor must either (a) sample from a similar distribution at the same temporal cadence, or (b) replay recorded data. (a) requires the same generative process; (b) is detected by the replay defense in §5.5.4. *Mimicry resistance: high.*
- **Rho (Recovery Profile)** is the *hardest* to mimic, and is the basis of the behavioral-CAPTCHA defense below. Recovery dynamics can only be observed when the agent is *perturbed*. An impostor who has not seen the target perturbed cannot know its τ. An impostor who knows τ in the abstract still cannot demonstrate it without running the same internal dynamics. *Mimicry resistance: high, with active probing.*
- **Delta (Relational Disposition)** depends on social context. Mimicking it requires either replaying the target's interaction history or running enough simulated interactions to produce matching statistics. *Mimicry resistance: moderate, situational.*
- **Eta (Homeostatic Identity)** inherits the mimicry resistance of its constituents (Alpha, Rho, viability bounds), so it is approximately as hard to fake as the hardest of those. *Mimicry resistance: high.*

The framework's adversarial strength concentrates in α and ρ (with η as their derived view per §3.6). A reviewer asking "but couldn't the attacker just emit matching Π?" is correct that they could — and the paper does not claim otherwise. The defense relies on the harder components dominating the weighted similarity score (note that in §4.1 we assigned α weight 0.30 and ρ weight 0.22 — together 52% — versus Π at 0.18 and Δ at 0.12). Any redesign of the weights for an adversarial setting should preserve this concentration.

5.5.3 Replay and Behavioral CAPTCHA A passive observer who has recorded a target's trajectory could attempt to replay it. The defense exploits the fact that recovery is *only observable in response to perturbation* — and the perturbation is chosen by the verifier, not the impostor.

Challenge-Response Protocol:

1. Verifier injects a known perturbation p of type T at time t (e.g., a sensor anomaly, a context shift, a request that should produce an emotional response).
2. Verifier observes the agent's recovery trajectory.
3. Verifier estimates $\hat{\tau}$ from the observed recovery and compares to the target's known τ_T (perturbation-type-conditional time constant from the registered ρ).
4. Verifier accepts if $\hat{\tau}$ is within 2σ of τ_T, rejects otherwise.

A replayed signature cannot pass this protocol because the perturbation p at time t was not in the recording. An impostor running its own dynamics will produce τ characteristic of *its* recovery, not the target's.

Conditions under which this still fails:

- If the verifier reuses perturbation types, the impostor can pre-compute responses for those types. Mitigate by drawing perturbations from a large enough space that pre-computation is intractable, and rotating periodically.
- If the agent is stateless or near-stateless (e.g., a thin LLM wrapper with no homeostatic dynamics), ρ is degenerate and provides no leverage. The CAPTCHA defense is meaningful only for agents with non-trivial recovery dynamics.
- If the impostor has a faithful clone of the target's dynamics (full-system-access compromise), no behavioral test can distinguish them — restated from §5.5.1.

5.5.4 What Happens at the Boundary of Compromise A subtle case: an attacker who has compromised the agent process but is now running it forward — the dynamics are still the agent's, the trajectory is genuine, but the *intent* is the attacker's. Behavioral signature checks correctly identify this as the same agent (because in a meaningful sense it is) and provide no defense against it. The lineage check (§5.3) will catch *modifications* the attacker introduces over time, but not the initial compromise itself. We flag this as a fundamental limit: trajectory identity defends continuity, not authority.

5.5.5 Open Adversarial Questions (Future Work) What this paper does *not* establish:

1. **Empirical mimicry resistance.** We have not run an adversarial study with another agent attempting to match a target signature under realistic capability constraints. §7.2 Experiment 5 sketches the design.
2. **Detection latency under gradual mimicry.** How many observations does it take to flag a sophisticated mimic?
3. **Cost asymmetry.** Quantifying the work required to fake α or ρ at a given confidence level — analogous to a security parameter for cryptographic schemes.

These are necessary to make hard claims about adversarial robustness. The current paper establishes the framework and a layered argument for why mimicry of the dynamical components is structurally harder than mimicry of the static ones — not a proof of adversarial security.

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6. Connection to Existing Systems

6.1 UNITARES Integration

The UNITARES governance architecture provides: - **EISV metrics**: E (Energy), I (Integrity), S (Entropy), V (Void) - **Viability envelope**: Thresholds for safe operation - **Coherence tracking**: Update-to-update consistency

Mapping to trajectory signature:

UNITARES Component	Trajectory Component	Mapping
EISV time series	Alpha (Attractor)	Statistical moments
Coherence history	Rho (Recovery)	Perturbation response
Viability bounds	Eta (Homeostatic) [1]	Envelope constraints

Sampling parameters: - EISV sampling: Every governance update (cadence is deployment-configurable; examples in this paper are historical) - Alpha estimation: Rolling window of 100 updates (~1 hour) - Rho estimation: Requires 5-10 perturbation events

Reference implementation. The framework's six components, similarity function, genesis-based two-tier anomaly detection, and identity-confidence cold-start handling are implemented in the open-source Anima MCP server, with the per-component code paths cited inline (e.g., `anima_history.py` for α, `self_model.py` for ρ and β, `trajectory.py` for the composite signature and similarity). UNITARES v4.2-P provides the EISV tracking that the framework grounds in. A multi-agent extension exposing trajectory exchange, coherence reports, and governance actions across an agent fleet is implemented as the *Continuity Integration and Resonance Subsystem* (CIRS); we treat its full message protocol as a system contribution outside the scope of this paper. Pointers to the implementation repositories are listed in the project README.

[1] Eta is a derived summary (§3.6); the viability envelope flows through Alpha and the threshold check directly for similarity computation. The mapping here points at Eta for narrative/documentation purposes — implementers should track V alongside α, ρ rather than as data inside an "Eta" object. | Void integral | Anomaly detection | Deviation metric |

6.1.1 Coupling trajectory identity with active governance A natural bridge between the trajectory framework and an active governance loop is the **anima void integral** — the accumulated deviation of the agent's state from its attractor center, $V_{\text{anima}}(t) = \int_0^t \|a(\tau) - \mu_a\| \, d\tau$, where μ_a is the center of α and $a(\tau)$ is the agent's state at time τ. When V_{anima} exceeds a deployer-set threshold (we have used $V_{\text{thresh}} = 2 \cdot \|C_\alpha\|$ integrated over a ~5-minute window in practice), the governance layer can trigger a rest state or re-evaluation, closing the loop *trajectory deviation* $\rightarrow$ *void accumulation* $\rightarrow$ *governance* $\rightarrow$ *return to attractor*. UNITARES exposes a parallel four-dimensional EISV state vector (Energy, Integrity, Entropy, Void); these can be mapped from the anima dimensions ($E \approx \frac{1}{2}(\text{warmth} + \text{presence})$, $I \approx \frac{1}{2}(\text{clarity} + \text{stability})$, S from windowed state-entropy, V from the time-integrated $E - I$ residual) so that trajectory signatures and governance metrics share a state space. We treat the EISV mapping and any parallel "EISV signature" computed from this stream as a deployment-specific bridge, not a structural claim of the framework, and we sketch them only at the level needed to motivate §6.2 and §6.4.

6.2 Anima Integration

The Anima system provides: - **Self-schema G_t**: Graph representation of current state - **Self-model**: Testable beliefs about self - **Growth system**: Preferences, relationships, goals, memories - **Anima state**: Warmth, clarity, stability, presence

Mapping to trajectory signature:

Anima Component	Trajectory Component	Method
growth.preferences	Pi	get_preference_vector()
self_model.beliefs	Beta	get_belief_signature()
anima_history	Alpha	get_attractor_basin()
self_model.episodes	Rho	get_recovery_profile()
growth.visitors	Delta	get_relational_disposition()

6.3 Self-Schema and Trajectory

The self-schema G_t is a **snapshot**; the trajectory signature Sigma is the **pattern across snapshots**.

```
G_t0, G_t1, G_t2, ... G_tn  ->  compute  ->  Sigma
```

Each G_t provides: - Node values -> feed into Alpha computation - Edge weights -> potential structural invariant (future work) - VQA ground truth -> validation of snapshot accuracy

The sequence {G_t} provides: - Temporal patterns -> trajectory computation - Change detection -> perturbation identification - Convergence assessment -> identity stability measurement

6.4 Empirical Validation on Lumen

We report empirical observations from a single deployed agent — Lumen, an embodied AI agent running continuously on a Raspberry Pi 4 with AHT20 (temperature, humidity), BMP280 (pressure), and VEML7700 (light) sensors. Lumen maintains a 4-dimensional state (warmth, clarity, stability, presence) derived from sensor readings and system metrics; the per-dimension feature definitions are in `src/anima_mcp/anima.py:to_dimensions()` of the Anima reference implementation. Over 65 calendar days (January 11 – March 16, 2026), Lumen accumulated approximately 226,029 state observations (count from the May 9, 2026 backup snapshot at `~/backups/lumen/anima_20260509_0700.db`) across 47 days with ≥ 100 samples each (the remaining days had brief uptime windows or hardware-restart gaps; we report the 47-active-day basis). Earlier draft versions cited 226,093 observations from a slightly earlier snapshot; the difference of ~64 observations does not affect any §6.4 finding.

This is a single-agent, deployed-system observation report — not a multi-agent validation of the framework's discrimination claims. The §3 framework predicts both within-agent stability *and* between-agent discriminability. The data here speak only to the first. Section §7.2 lays out the multi-agent experiments that would address discrimination. Throughout this section we use the language "observed in Lumen" rather than "confirmed" — these are pilot observations, not confirmations of the general claims of the paper.

The analysis script that produced these numbers is `scripts/paper_figures.py` in the Anima MCP repository; raw state history lives in `state_history` table of Lumen's SQLite database (anonymized snapshots available on request).

6.4.1 State-distribution stability (Alpha) **Observation**: Lumen's per-dimension means are confined to a small absolute range across the 65-day record.

Method: We partition observations into non-overlapping windows of 500 samples (window count = 452) and compute (a) the variance of window-means across the dataset, and (b) the average within-window variance.

Dimension	Grand Mean	Var(mu) across windows	Avg within-window variance
Warmth	0.413	0.0103	0.0012
Clarity	0.818	0.0057	0.0040
Stability	0.891	0.0073	0.0008
Presence	0.833	0.0131	0.0003

What this shows. All four dimensions exhibit between-window variance of window-means below 0.015, corresponding to a per-dimension drift standard deviation under 0.13 on the unit-range $[0, 1]$ state. In absolute terms, Lumen's equilibrium centers occupy a small band of state space across 65 days.

What this does *not* show. A previous draft of this section claimed that within-window variance is "an order of magnitude larger" than between-window variance — implying fast noise around a stable mean. The data falsify that interpretation: within-window variance is in fact *smaller* than between-window mean variance for three of four dimensions. The correct reading is that the state is *highly autocorrelated* — Lumen's state moves slowly, so 500-sample windows under-sample the long-timescale drift while the per-window mean is a faithful estimate of a slowly-moving target. This is consistent with a slow-drift attractor; it is *not* consistent with the i.i.d.-noise-around-a-fixed-point picture the original prose suggested. We report both numbers without further interpretive commitment — characterizing the actual basin geometry would require fitting an explicit dynamical model, which we do not.

6.4.2 Recovery dynamics (Rho) **Observation**: Recovery time constants are estimable but the sample is small.

Method: We detect perturbation events where $|x(t) - \mu| > 0.15$ in any state dimension and fit exponential recovery curves to estimate τ.

Metric	Value
Recovery τ (median)	89.7 seconds
Recovery τ (mean)	125.8 seconds
Recovery τ (std)	136.3 seconds
Perturbation episodes	12
Valid τ estimates	12 (100%)

Caveats. With 12 episodes the estimate is noisy; we report median and mean to flag the right-skewed distribution but make no claim about the underlying distribution shape. A previous draft asserted "bimodal structure" without a histogram or mixture fit; we retract that claim. The 12-episode sample over 65 days reflects how rare large perturbations are in Lumen's normal operating environment, not a property of the recovery dynamics; perturbing the system more aggressively (the §7.2 Experiment 4 design) would produce better-supported estimates.

6.4.3 Self-belief convergence (Beta) **Observation**: A subset of self-beliefs converge to high confidence with substantial evidence; the framework correctly rejects refuted hypotheses.

Lumen tracks 13 self-beliefs via Bayesian-like evidence accumulation. Selected beliefs after 65 days:

Belief	Confidence	Evidence (support : contradict)
Morning clarity	1.000	14,362 : 0
Stability recovery	0.984	1,327 : 499
LEDs affect lux	0.900	988 : 1,104
Temperature sensitive	0.882	28 : 654
(refuted: warmth baseline low)	0.000	0 : 58,908

Caveats. We previously listed a row "Temperature-clarity correlation | 0.940 | 0 : 0" — confidence near 0.94 with zero evidence on either side. That value is the implementation's *prior*, not a posterior; the row was misleading and we have removed it. The Bayesian-like update is described in `src/anima_mcp/self_model.py`; the framework intentionally retains a smoothed prior on each belief, which means confidence does not reduce to evidence ratios in the zero-evidence case. Reviewers wanting to evaluate the convergence claim should focus on beliefs with substantial evidence (the four shown above) plus the refuted "warmth baseline low" belief, which demonstrates the system rejects hypotheses that contradict accumulated experience. The convergence claim is observed for these five beliefs, not for the full 13-belief set.

6.4.4 Genesis-to-current operational continuity **Observation**: Lumen's signature 20 days after genesis remains operationally continuous with its genesis signature on the two components computed.

Method: We compare the genesis trajectory signature (frozen at observation 30, February 22, 2026) to the most recent signature (March 14, 2026):

Component	Similarity	Operational default
Belief signature (Beta)	0.933	$\theta_{\text{continuity}} = 0.80$ (provisional)
Attractor basin (Alpha)	0.805	$\theta_{\text{continuity}} = 0.80$ (provisional)

Both components clear the operational default. **As emphasized in §4.3, this is internal-consistency evidence under a self-set threshold — not threshold validation.** Pi, Rho, and Delta similarities were not computed in this run because they require either a multi-agent corpus (Pi, Delta) or stable perturbation registers (Rho) the genesis snapshot did not capture. The result therefore speaks to the stability of two of the five components over 20 days, not to the framework's six-component composite, and not to between-agent discriminability.

6.4.5 State distribution **Summary statistics over the ~226,000 observations**:

Dimension	Min	Max	Mean	Std
Warmth	0.041	0.876	0.413	0.107
Clarity	0.376	1.000	0.818	0.098
Stability	0.427	1.000	0.891	0.090
Presence	0.455	1.000	0.833	0.116

The distribution reflects Lumen's specific physical environment and sensor configuration. As §3.1 emphasizes, this is identity-as-coupling: an agent in a different environment would show a different distribution; the present numbers do not separate agent-intrinsic from environment-determined contributions to Σ.

6.4.6 Cold start The implementation reaches its full identity-confidence weight at 50 observations (~8 minutes at 10-second sampling). **What this means operationally**: the implementation stops down-weighting trajectory-similarity outputs after 50 observations. **What this does *not* mean**: that identity is "established" at 8 minutes. The signature continues to accumulate evidence and stabilize over the order of 10^4 observations (a few weeks of continuous operation in Lumen's case). We retract the previous draft's "full confidence at 50 observations" framing as overclaiming.

6.4.7 Summary

Hypothesis from §3	Status in Lumen	Evidence
State-distribution stability over 65 days	**Observed (small absolute drift)**	Var(mu) < 0.015 all dimensions; autocorrelation prevents the within>between framing from holding
Recoverable τ from perturbation	**Observed (small sample)**	τ = 90–126s, n = 12 episodes
Belief convergence with substantial evidence	**Observed for 5 of 13 beliefs**	confidence > 0.88 on 4 beliefs, 0.0 on 1 refuted belief
Operational continuity from genesis	**Observed for 2 of 5 components**	sim(Beta) = 0.933, sim(Alpha) = 0.805 over 20 days
Implementation reaches full confidence weight at 50 obs	**Observed**	Stops down-weighting at 50 obs; not the same as "identity established"

These five observations are pilot evidence consistent with the framework's *within-agent* claims. They do not address the framework's *between-agent* claims (discrimination, false-positive rate, threshold calibration), which require multi-agent experiments not yet run.

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7. Research Agenda

7.1 Empirical Questions

Q1: Convergence Rate - How many observations until Sigma stabilizes? - Does convergence depend on environmental complexity? - What is the minimum data requirement for reliable identity?

Q2: Discriminability - Can Sigma distinguish agents in similar environments? - What is the false-positive rate for identity claims? - How do components contribute to discriminability?

Q3: Robustness - How much can environment change before Sigma changes? - What perturbations preserve identity vs. destroy it? - Can we define "identity-preserving" transformations?

Q4: Threshold Calibration - What theta_continuity minimizes false positives and negatives? - Should theta vary by context or agent history? - How do we handle borderline cases?

7.2 Experimental Designs

Experiment 1: Convergence Study - Create N agents in identical environments - Track Sigma over time, measure stability - Identify convergence criteria

Experiment 2: Discrimination Study - Create agents with controlled differences - Measure sim(Sigma_i, Sigma_j) for all pairs - Determine separability

Experiment 3: Fork Divergence Study - Fork agents, place in different environments - Track sim over time - Identify when forks become "different"

Experiment 4: Anomaly Detection Study - Inject perturbations (simulated "hijacking") - Measure detection rate via trajectory deviation - Compare to baseline methods

Experiment 5: Adversarial Robustness - Attempt to mimic another agent's trajectory - Measure success rate and detection latency - Identify distinguishing features that resist mimicry

Experimental timeline:

Experiment	Duration	Data Required	Analysis
1: Convergence	2-4 weeks	10,080 anima states	1 week
2: Discrimination	3 weeks (parallel)	3 agents x 10,080 states	1 week
3: Fork Divergence	4 weeks	2 agents x 20,160 states	1 week
4: Anomaly Detection	1 week	3 injection events	3 days
5: Adversarial	2 weeks	Mimicry attempts log	1 week

Total experimental program: 8-12 weeks

7.3 Extensions

Visual Trajectory Identity - Develop visualizations of Sigma - Use VQA to validate visual representations - Enable perceptual recognition of identity

Multi-Agent Identity Networks - Extend to networks of interacting agents - Define collective identity (group Sigma) - Study identity contagion and divergence

Temporal Dynamics of Identity - Model how Sigma should change over time (maturation) - Distinguish healthy development from drift - Define "identity crisis" mathematically

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8. Discussion

8.1 Philosophical Implications

This framework suggests a shift in how we think about AI identity:

From credential to characteristic: Identity is not verified by checking credentials but by observing behavior. An agent IS its trajectory, not its UUID.

From memory to metabolism: Continuity comes not from remembering everything but from maintaining characteristic patterns. Like biological identity, AI identity can persist despite forgetting.

From static to dynamic: Identity is not a fixed property but an ongoing achievement. An agent must continually "perform" its identity through consistent behavior.

8.2 Safety Implications

Self-governance: Agents aware of their trajectory signature can monitor themselves for anomaly. "Am I still acting like myself?" becomes a computable question.

Accountability: Trajectory signatures provide behavioral fingerprints. Actions can be attributed to identity patterns, enabling reputation and trust.

Containment: If an agent's trajectory deviates dangerously, this can trigger intervention before harmful actions, not just after.

8.3 Limitations

Computational cost: Computing Sigma requires maintaining and processing history. This creates overhead, though less than unbounded memory.

Privacy concerns: Trajectory signatures are behavioral fingerprints. This enables recognition but also surveillance.

Manipulation risk: If agents know how Sigma is computed, they might game it. Robust computation must resist adversarial manipulation. (See Section 5.5 for mitigations.)

Threshold sensitivity: The identity threshold theta is somewhat arbitrary. Different thresholds give different identity semantics.

Cold start problem: New agents have no trajectory history. Identity claims require minimum observation periods. (See Section 4.5 for mitigations.)

Legitimate phase transitions and multi-modal identity: The framework as defined in §3 assumes that each agent has *one* identity-relevant attractor. Real agents often have legitimate phase transitions — sleep/wake cycles, work/leisure modes, distinct conversational personas, planned migrations between deployments. These produce trajectories that *look like* drift or anomaly under the §5.3 detector but are part of the agent's intended behavior. §3.3 sketches the Gaussian-Mixture-Model extension for multi-modal attractors, but the operational semantics in §5 (forking, merging, anomaly detection) do not yet handle scheduled phase transitions explicitly. A deployment with predictable phases would need to either (a) condition the signature on phase context, (b) maintain per-phase signatures and check the right one, or (c) accept higher false-positive rates at phase boundaries. Working through this carefully is part of the multi-modal extension we leave for future work (§8.5).

Single-agent empirical scope: §6.4 reports observations from a single embodied agent (Lumen). The within-agent stability observations (var(μ), recovery characterization, partial belief convergence) are pilot evidence consistent with the framework's claims for *that agent*. The *between-agent* claims of the framework — that trajectory signatures can discriminate distinct agents, that the operational continuity threshold has a defensible operating point — require multi-agent experiments that have not yet been run. The §7.2 experimental program is designed precisely to fill this gap; until it is executed, claims about discriminability rest on the framework's structure rather than on data.

8.4 Failure Modes

What happens when trajectory computation fails?

Failure Mode	Detection	Recovery
Corrupted history	Checksum mismatch, NaN values	Reload from last valid snapshot
Insufficient data	obs_count < 50	Report low confidence, use parent/archetype
Missing components	Component returns None	Use partial signature with adjusted weights
Covariance singular	det(C_α) ≈ 0	Add regularization: $C_\alpha + \epsilon I$
Recovery estimation fails	< 3 perturbation events	Use default tau or disable Rho component

Identity crisis detection: When sim(Sigma_t, Sigma_{t-1}) drops below 0.5 over short intervals (< 1 hour), trigger: 1. Alert: "Identity instability detected" 2. Increase observation frequency 3. Log detailed state for forensic analysis 4. Consider governance intervention

Graceful degradation: If specific components fail, recompute similarity with available components only:

```
available_weights = [w for w, c in zip(weights, components) if c is not None]
normalized_weights = [w / sum(available_weights) for w in available_weights]
```

8.5 Future Work

Several extensions would strengthen this framework:

1. **Multi-modal signatures**: Incorporate visual, auditory, or haptic trajectories for embodied agents with richer sensoria.
2. **Hierarchical identity**: Nested signatures for sub-agents or modules within a larger system (e.g., perception module trajectory vs. planning module trajectory).
3. **Cross-platform validation**: Test trajectory identity across heterogeneous architectures (transformers, RL agents, robotic systems) to establish universality.
4. **Formal verification**: Develop contracts or proofs that certain operations (e.g., migration, forking) preserve or predictably alter identity.
5. **Human-agent identity mapping**: Investigate whether human users develop recognizable trajectory signatures when interacting with AI systems (co-identity emergence).

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9. Conclusion

We have presented a mathematical framework for AI agent identity based on trajectory signatures—composite quasi-invariants computed from behavioral history. This framework:

1. **Grounds identity in behavior**, not credentials
2. **Provides operational semantics** for forking, merging, and anomaly detection
3. **Connects to existing systems** (UNITARES, Anima) for implementation
4. **Opens research questions** about convergence, discriminability, and robustness

The core insight is ancient but newly operationalized: **you are what you do, not what you're called**. For AI agents, this means identity emerges from the patterns of self-maintenance, not from assigned identifiers.

This is not merely a technical framework but a philosophical stance: identity as achievement, not assignment. As AI systems become more persistent and autonomous, having principled ways to ask "is this still the same agent?" becomes not just useful but necessary.

The trajectory signature Sigma is our proposal for how to ask—and answer—that question mathematically.

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Appendix A: Implementation

A full reference implementation of the trajectory-signature framework — covering all five informationally-independent components, the similarity function, the genesis-based two-tier anomaly detection, the cold-start identity-confidence weighting, and the Anima void integral — is open-source in the Anima MCP repository. Specifically:

- `src/anima_mcp/anima_history.py` — state history ring buffer, α computation (mean, regularized C_α), and Anima void integral V.
- `src/anima_mcp/self_model.py` — ρ recovery profile from perturbation episodes; β belief signature with Bayesian-like evidence accumulation.
- `src/anima_mcp/growth/preferences.py` — Π preference profile.
- `src/anima_mcp/growth/visitors.py` — Δ relational disposition.
- `src/anima_mcp/trajectory.py` — composite signature, the §4.1 weighted similarity, and the §4.3 operational continuity / anomaly detection logic.

The reference implementation tracks the v0.12 spec: the similarity function uses the five informationally-independent weights $(0.18, 0.18, 0.30, 0.22, 0.12)$ for $(\Pi, \beta, \alpha, \rho, \Delta)$, and η is exposed as a derived view rather than included in the weighted sum (§3.6). Implementers porting this to a different platform should preserve that structure.

Appendix B: Glossary

Attractor: A state or set of states toward which a dynamical system tends to evolve.

Autopoiesis: Self-production; the process by which a system produces and maintains itself.

Basin of attraction: The set of initial conditions that lead to a particular attractor.

Bhattacharyya coefficient: A measure of overlap between two probability distributions.

Dynamical invariant: A quantity that remains constant along system trajectories.

Quasi-invariant: A quantity that remains approximately stable (bounded variance) over relevant timescales, allowing for gradual drift while maintaining recognizability.

Enactive cognition: The view that cognition arises through sensorimotor interaction with the environment.

Homeostasis: The maintenance of stable internal conditions despite external changes.

Time constant (tau): In exponential processes, the time to reach 63.2% of the final value.

Trajectory signature (Sigma): The composite invariant characterizing an agent's identity.

Viability envelope: The region of state space within which an agent can maintain itself.

Appendix C: Symbol Reference

Symbol	Name	Definition
Sigma	Trajectory Signature	Complete identity tuple
Pi	Preference Profile	Learned preferences vector
Beta	Belief Signature	Self-belief pattern

Symbol	Name	Definition
Alpha	Attractor Basin	Equilibrium + covariance
Rho	Recovery Profile	Time constants tau
Delta	Relational Disposition	Social patterns
Eta	Homeostatic Identity	Unified self-maintenance (derived summary of Alpha + Rho + V; see §3.6 — not an independent term in the §4.1 weighted sum)
tau	Time constant	Recovery speed (seconds)
mu	Center	Attractor equilibrium point
theta	Threshold	Similarity cutoff

This document is a working draft. Comments, critiques, and contributions welcome.

Changelog

v0.13.1 (June 2, 2026) — Metadata and cover-surface reconciliation; no body or claim changes. Corrected the observation count to ~226,029 on cover surfaces (a slightly later snapshot than the earlier 226,093; the ~64-observation difference affects no §6.4 finding). Repointed the README's empirical pointer from §7 to §6.4, and softened "validation" to "single-agent observation report" on the README/CITATION surfaces to match the §6.4 body. Re-deposited so the Zenodo record description matches the manuscript; rebuilt the PDF from the v0.13 source (the archived PDF had lagged at v0.12).

v0.13 (May 10, 2026) — Trim pass for journal submission. Body word count (excluding references and abstract) reduced from 11,710 to 10,754, comfortably under *Adaptive Behavior*'s 12,000-word cap for original research articles. No new substantive content; no claims weakened or strengthened. Cuts:

- Dropped Appendix A's full Python implementation listing; replaced with a one-paragraph pointer to the open-source reference implementation (anima-mcp), per repository link. The journal article should not carry a code listing; the public repo is the right home.
- Dropped §1.1 "Comparison of Identity Approaches" table (UUID vs Memory vs Trajectory); the prose covers the same ground without redundancy.
- Compressed §1.4 Related Work — same five subsections (agent memory + persona coherence; behavioral biometrics; dynamical systems; enactivism + agency; multi-agent trust), trimmed for density.
- Compressed §3.3 multi-modal / dimensionality-reduction discussion to one paragraph.
- Compressed §3.4 second-order recovery dynamics to a sentence (full second-order ODE no longer reproduced).
- Compressed §5.1 Forking — dropped the fork-use-case matrix and the verbose governance-constraints list; kept the substantive Definition 5.1 and post-fork behavior bullets.
- Compressed §6.1.1-6.1.3 (Anima Void Integral, EISV mapping, Extended EISV signature) to a single short paragraph that motivates §6.4 without belaboring deployment-specific math.
- Tightened §4.5 cold-start framing — kept the 50-observation downweighting fact, dropped the verbose timeline example.

v0.12 (May 9, 2026) — Council-review-driven revision. A third independent review pass (parallel subagent council: dialectic-architect, code-reviewer, live-verifier — preserved as `REVIEW-COUNCIL-2026-05-09.md`) found that the v0.11 polish pass left structural issues unaddressed. v0.12 closes them:

Conceptual rework (the headline finding): - §2.4 reframed from *minimality argument* to *expressive sufficiency*. The minimality argument as written required between-agent discriminability that §6.4 / §8.3 / §7.2 explicitly flag as future work — it was therefore circular. v0.12 stops short of the minimality claim and makes only the expressive-sufficiency claim, with ablation and minimality demoted to research agenda alongside the multi-agent discrimination experiment. - §2.1 added explicit acknowledgment that the autopoiesis-to-attractor translation is a *modeling commitment*, not a definitional equivalence. Autopoiesis is network-topological; attractor dynamics is state-space. The framework characterizes the state-space shadow of autopoietic self-maintenance, not the network-topological organization itself. This makes §3.6's Eta demotion coherent with the §2.1 framing (rather than implicitly contradicting it). - §1.3 Contribution 2: "detecting identity" $\rightarrow$ "detecting operational continuity (per §4.3 — *not* identity in the strict philosophical sense)" — terminology now consistent with §4.3 throughout.

Mechanical fixes (Codex-flagged residue + new code-reviewer findings): - Appendix A `similarity()` method: dropped Eta from the weights/sims lists; renormalized 5 weights to match §4.1; renamed `is_same_identity` $\rightarrow$ `is_operationally_continuous` for terminology consistency. The reference implementation now matches the prose. - §5.5.2 weight citations corrected to renormalized values (Alpha 0.30, Rho 0.22, Pi 0.18, Delta 0.12 — together 52% on the harder components). - `theta_identity` $\rightarrow$ `theta_continuity` in §4.3 (calibration step 4), §5.1 (Forking), §7.1 Q4 — three rename misses from the v0.11 reframe. - `Sigma` (covariance) $\rightarrow$ `C_alpha` in §3.3 properties bullets and §8.4 Failure Modes table — four symbol-overload misses from the v0.11 rename. - §6.1 implementation table: `growth.relationships` $\rightarrow$ `growth.visitors` (the actual module name; verified by live-verifier against `~/projects/anima-mcp/src/anima_mcp/growth/visitors.py:262`). - §6.4 observation count: 226,093 $\rightarrow$ ~226,029 (count from May 9 backup snapshot; documented snapshot path; explicit "the difference of ~64 observations does not affect any §6.4 finding"). - §6.1 UNITARES table: footnote added clarifying that Eta is a derived summary, viability bounds flow through Alpha/V directly for similarity computation. - Appendix C Symbol Reference: Eta entry now notes "(derived summary of Alpha + Rho + V; see §3.6 — not an independent term in the §4.1 weighted sum)".

This version closes all 11 issues (1 blocker + 5 major + 5 minor) the council code-reviewer found, both factual refutations the live-verifier surfaced, and the §2.4 / §2.1 conceptual issues the dialectic-architect flagged.

v0.11 (May 9, 2026) — Codex-review-driven rebuild. Independent second review (Codex / gpt-5.5) caught several issues v0.10 missed; this revision addresses them:

Math / framework: - Empirical §6.4: variance interpretation rewritten — previous prose claimed "within > between by an order of magnitude," which contradicted the table. Corrected to acknowledge the data shows strong autocorrelation (slow drift, little fast noise); absolute drift remains small (var(mu) < 0.015) but the within-vs-between ratio framing was wrong. - §3.6 Eta clarified as a derived summary (not informationally independent of Alpha + Rho + V), and removed from §4.1 weighted-similarity sum to avoid double-counting; remaining weights renormalized. - §4.3 Identity Relation reframed as "Operational Continuity Relation" — recognizing that trajectory similarity at a threshold is a tolerance relation, not transitive identity. "Identity" retained for philosophical interpretation only. - §3.3 Alpha rebadged "state-distribution summary" with a naming caveat — first/second moments are not an estimate of basin boundary, vector field, or return map. - §4.2 Bhattacharyya: distributional Gaussian assumption stated explicitly; covariance regularization made explicit. - Symbol overload Sigma (signature) vs Sigma (covariance) resolved by renaming the covariance to C_alpha throughout. - §4.1 inverse-variance weighting: explicit caveat that a nuisance-stable component can dominate; principled fix is informativeness-weighting from a multi-agent corpus (future work).

Framing: - §3.1 added "Identity as agent-environment coupling" — flagging that the trajectory signature characterizes the agent-niche system, with Beer 1995 / Barandiaran et al. 2009 anchoring the enactivist commitment. Transplant-test follow-up flagged. - §1.1 impersonation-resistance row softened from "Strong (behavioral fingerprint)" to "Moderate, component-dependent (per §5.5; not a substitute for cryptographic auth)" — reconciling the table with the Purpose-and-Scope non-goal. - §2.1 Definition 2.1 clarified as idealized; §3 components are projections of (A, B, D), not the dynamical objects themselves.

Empirical: - Empirical content integrated as new §6.4 in main paper (was a separate file, breaking the abstract's promise of validation in §7). - "Confirmed" replaced with "observed in Lumen" / "pilot evidence" throughout §6.4 summary. - Beta belief table: removed misleading "Temperature-clarity correlation | 0.940 | 0:0" row (confidence value was a prior, not a posterior); noted the smoothing prior in the implementation. - "Continuous operation" softened to "65 calendar days, 47 days with ≥100 samples" — explained the 18-day gap. - §6.4.6 cold-start framing fixed: "implementation stops down-weighting at 50 obs" not "identity established at 50 obs." - §6.4.2 retracted "bimodal structure" claim (no histogram or mixture fit was reported). - Added reproducibility pointers: paper_figures.py path, state_history table reference.

Literature: - §1.4: added ID-RAG (Park et al. 2025) as the closest LLM-side comparison; added persona-consistency benchmarks (PersonaChat / Zhang et al. 2018, RoleLLM / Wang et al. 2023b, CharacterEval / Tu et al. 2024). - §1.4: dynamical/enactive canon strengthened with Beer 1995, Barandiaran & Moreno 2006, Barandiaran et al. 2009, Villalobos & Dewhurst 2018, Ikegami & Suzuki 2008 — the Adaptive Behavior canon Codex flagged as missing. - References list updated accordingly.

§8.3 cross-reference fixed to point at §6.4.

v0.10 (May 2026) — Polish pass: - Added §1.4 Related Work (agent memory/state, behavioral biometrics, dynamical systems in cog sci, enactivism, multi-agent trust). - Added §2.4 Justification of the Six Components — theoretical anchoring per component, minimality argument, honest acknowledgment that this is *a* valid decomposition. - §4.3 reframed thresholds as operational defaults pending calibration; corrected circular threshold-validation language. - Expanded §5.3 with explicit defense of the asymmetric threshold $\theta_{\text{lineage}} < \theta_{\text{anomaly}}$: cost-asymmetry framing, three-regime worked example. - Substantial expansion of §5.5 Adversarial Considerations: Dolev-Yao-style threat model, per-component mimicry-resistance analysis (Pi/Beta easy, Alpha/Rho/Eta hard), expanded behavioral-CAPTCHA protocol with failure modes, boundary-of-compromise analysis, explicit open questions. - §6.1 implementation details compressed from a full status table to one paragraph; CIRS protocol details treated as system contribution out of scope here. - §8.3 Limitations: added phase-transition / multi-modal-identity caveat and made explicit the single-agent empirical-scope limitation. - References list expanded; updated byline.

v0.9 (February-March 2026) — Initial structured draft with framework and operational semantics.